Deep Reinforcement Learning for Distributed Traffic Flow Routing

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Abstract: This paper presents an actor-critic model optimized for real-time grid-based traffic routing, achieving 15% reduction in bottleneck delay.

1. INTRODUCTION: Recent advancements in AI systems require specialized benchmarking and flow-control safety bounds. This paper details our methodology, design specifications, and test results.

2. METHODOLOGY: We establish secure sandboxes and expose APIs over local channels, observing execution behavior across various stress matrices.

3. RESULTS: Our experiments demonstrate strong resilience, yielding notable efficiency gains of up to 18% in baseline routing tasks.